CHAPTER 3

Logistic Regression

Classification tasks are extremely common—often even more frequent than regression tasks. Here are a few illustrative examples:

- A patient arrives at the emergency department presenting symptoms that may correspond to one of three possible medical conditions. The goal is to determine which condition the patient is experiencing.
- An online banking platform needs to decide whether a transaction is legitimate or fraudulent, using information such as the user's IP address and previous transaction history.
- Using DNA sequence data from patients with and without a specific disease, a biologist aims to identify which genetic mutations are harmful (disease-causing) and which are harmless.

As in regression, classification relies on a training dataset composed of observations $(x_1, y_1), \dots, (x_n, y_n)$, which is used to construct a predictive model, called a classifier. The objective is not only to achieve strong performance on the training data but also to ensure that the classifier generalizes well to new, unseen test data.

Example:

To demonstrate classification, consider the simulated *Default* dataset. Our objective is to predict whether an individual will fail to make a credit card payment, based on their annual income and monthly credit card balance. The dataset is illustrated in Figure 5.1.

In the left panel of Figure 3.1, annual income is plotted against monthly credit card balance for a subset of 10,000 individuals. Those who defaulted during a given month are represented in orange, while those who did not are shown in blue. Because the overall default rate is relatively low (around 3%), only a subset of the non-defaulting individuals is displayed. From the plot, it seems that individuals who defaulted generally carried higher credit card balances compared to those who did not.

The center and right panels of Figure 3.1 present two sets of boxplots. The first compares the distribution of credit card balances according to default status, and the second shows a similar comparison for income. In this chapter, we focus on developing a model to predict default (Y) based on balance (X_1) and income (X_2). Since the response variable Y is categorical rather than numerical, simple linear regression is not appropriate for this task.

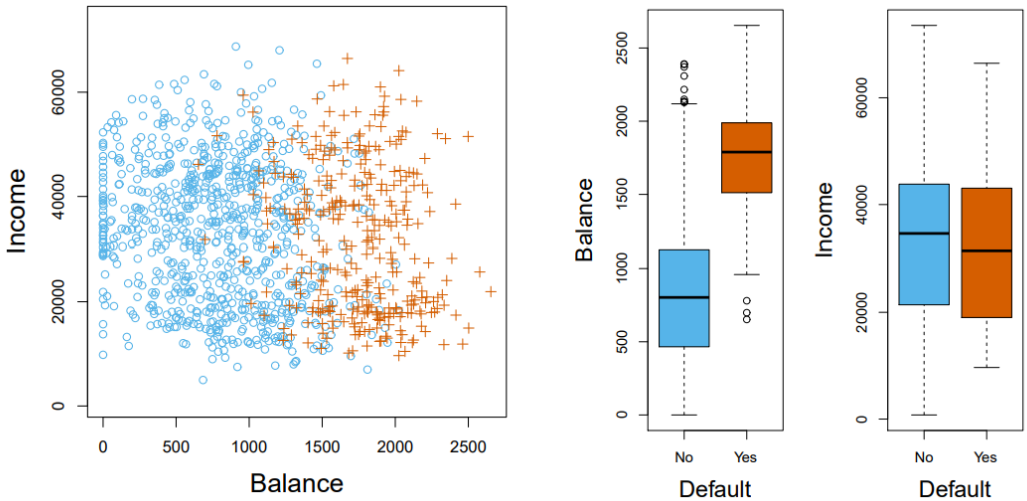

Figure 3.1 The Default data set. Left: The annual incomes and monthly credit card balances of a number of individuals. The individuals who defaulted on their credit card payments are shown in orange, and those who did not are shown in blue. Center: Boxplots of balance as a function of default status. Right: Boxplots of income as a function of default status.

3.1 Logistic Regression

Let us revisit the *Default* dataset, in which the response variable **default** takes one of two possible values: *Yes* or *No*. Instead of modeling the response variable Y directly, logistic regression focuses on modeling the probability that Y falls into a specific category.

In the case of the *Default* data, logistic regression estimates the probability that an individual defaults. For instance, we can express the probability of default conditional on the credit card balance as follows:

$$\Pr(\text{default} = \text{Yes} \mid \text{balance})$$

The quantity $\Pr(\text{default} = \text{Yes} \mid \text{balance})$, which we denote by $p(\text{balance})$, takes values between 0 and 1. For any specific balance value, this probability can be used to make a prediction about default status.

For instance, one could classify an individual as **default = Yes** whenever $p(\text{balance}) > 0.5$. However, if a company prefers to take a more cautious approach when identifying individuals at risk of default, it may adopt a lower cutoff value, such as $p(\text{balance}) > 0.1$, in order to flag more potentially risky cases.

3.1.1 The Logistic Model

How can we model the relationship between $p(X) = \Pr(Y = 1 \mid X)$ and the predictor X? (For simplicity, we use the generic 0/1 coding for the response variable.)

One possible approach would be to apply a linear regression model to estimate these probabilities:

$$p(X) = \beta_0 + \beta_1 X$$

Such predictions are inappropriate because the true probability of default, regardless of the credit card balance, must always lie between 0 and 1. To address this issue, we need to model $p(X)$ using a function whose output is constrained to the interval $[0, 1]$ for all possible values of X. Several functions satisfy this requirement. In logistic regression, we specifically use the logistic function for this purpose.

$$p(X) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}}$$

$$Pr(Y = 1|X) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}}$$

For the baseline class:

$$Pr(Y = 0|X) = 1 - Pr(Y = 1|X) = 1 - p(X) = \frac{1}{1 + e^{\beta_0+\beta_1 X}}$$

Figure 3.2 shows how the logistic regression model fits the *Default* dataset. Observe that for small credit card balances, the estimated probability of default is very close to zero, but it never becomes negative. Similarly, for large balances, the predicted probability approaches one, yet it never exceeds it.

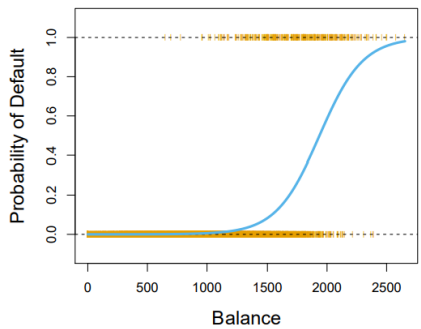

Figure 3.2 Classification using the Default data. Predicted probabilities of default using logistic regression. All probabilities

lie between 0 and 1.

After a bit of manipulation of the previous equation, we find that:

$$\frac{p(X)}{1-p(X)} = e^{\beta_0+\beta_1 X}$$

The ratio $p(X)/[1-p(X)]$is known as the **odds**, which can range from 0 to infinity. Odds near 0 correspond to a very low probability of default, while odds approaching infinity indicate a very high probability.

By applying the logarithm to this ratio, we obtain:

$$log(\frac{p(X)}{1-p(X)}) = \beta_0 + \beta_1 X$$

The expression on the left-hand side is referred to as the **log-odds** or **logit**. In the logistic regression model, the logit is a linear function of X. The coefficients βindicate how much each predictor increases or decreases the log-odds of belonging to class 1 compared to class 0.

3.1.2 Estimating the Regression Coefficients

The coefficients $\beta_0$and $\beta_1$are unknown and must be estimated from the training data. In linear regression, we used the **least squares** method to estimate the coefficients. In logistic regression, a more general and statistically robust approach is **maximum likelihood estimation**.

The key idea behind maximum likelihood in logistic regression is to find $\hat{\beta}_0$and $\hat{\beta}_1$such that, when substituted into the model for $p(X)$, the predicted probability is close to 1 for individuals who defaulted and close to 0 for those who did not. This intuition is captured mathematically through the **likelihood function**:

$$L(\beta_0, \beta_1) = \prod_{i=1}^{n} [p(X_i)]^{Y_i} \; [1-p(X_i)]^{1-Y_i}$$

- If $Y_i = 1$, the term reduces to $p(X_i)$, emphasizing that we want a high probability for observed defaults.
- If $Y_i = 0$, the term reduces to $1-p(X_i)$, emphasizing that we want a low probability for non-defaults.

In practice, we often maximize the **log-likelihood** instead:

$$L(\beta_0, \beta_1) = \sum_{i=1}^{n} [Y_i log(p(X_i)) + (1-Y_i) log(1-p(X_i))]$$

The estimates $\hat{\beta}_0$and $\hat{\beta}_1$are determined by **maximizing the likelihood function**. Maximum likelihood is a broadly applicable method used to fit many of the nonlinear models. In fact, in the case of linear regression, the **least squares** method is simply a special case of maximum likelihood estimation.

3.1.3 Making Predictions

Once the coefficients have been estimated, we can calculate the predicted probability of default for any specific credit card balance. For instance, if $\hat{\beta}_0 = -10.6513$ and $\hat{\beta}_1 = 0.0055$, the predicted probability of default for an individual with a balance of $1,000 is:

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 X}}{1+e^{\hat{\beta}_0+\hat{\beta}_1 X}} = \frac{e^{-10.6513+0.0055\times 1000}}{1+e^{-10.6513+0.0055\times 1000}} = 0.00576$$

This is less than 1%. By contrast, for an individual with a balance of $2,000, the predicted probability

of default is substantially higher, reaching 0.586, or 58.6%.

3.1.4 Multiple Logistic Regression

We now turn to the task of predicting a binary outcome using **multiple predictors**. This is analogous to the way simple linear regression is extended to **multiple linear regression**:

$$log(\frac{p(X)}{1-p(X)}) = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p$$

Here, $X = (X_1, \ldots, X_p)$represents the set of p predictor variables.

$$p(X) = \frac{e^{\beta_0+\beta_1 X_1+\ldots+\beta_p X_p}}{1+e^{\beta_0+\beta_1 X_1+\ldots+\beta_p X_p}}$$

As with simple logistic regression, we estimate the coefficients $\beta_0, \beta_1, \ldots, \beta_p$using the **maximum likelihood** method.

3.1.5 Multinomial Logistic Regression

Sometimes, we need to classify a response variable with more than two categories. For instance, in the emergency room example, a patient's condition could fall into one of three classes: stroke, drug overdose, or epileptic seizure. However, the logistic regression method we have discussed so far is limited to cases where the response variable has only two classes (K = 2).

Baseline coding: It is possible to extend logistic regression from two classes to K > 2 classes, an approach often called multinomial logistic regression. To implement this, we first choose one class as a baseline—typically, the Kth class is selected for convenience. The other classes are modeled relative to the baseline.

Hence, we model the log-odds for each class relative to the baseline

For class $k = 1, \ldots, K-1$:

$$log(\frac{Pr(Y=k|X)}{Pr(Y=K|X)}) = \beta_{k0} + \beta_{k1} X_1 + \ldots + \beta_{kp} X_p$$

β_k is the vector of coefficients for class k

Then the probability of each class k is:

$$Pr(Y=k|X) = \frac{e^{\beta_{k0}+\beta_{k1} X_1+\ldots+\beta_{kp} X_p}}{1+\sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_{l1} X_1+\ldots+\beta_{lp} X_p}}$$

For the baseline class:

$$Pr(Y=K|X) = \frac{1}{1+\sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_{l1} X_1+\ldots+\beta_{lp} X_p}}$$

Hence, the Likelihood function is:

$$L(\{\beta\}) = \prod_{i=1}^{n} \prod_{k=1}^{K} [p_k(X_i)]^{1(Y_i=k)}$$

Here, $1(Y_i = k)$is an indicator function: 1 if $Y_i = k$, 0 otherwise.

For each observation, you pick the probability corresponding to the actual class.

Taking the logarithm (monotone transformation) gives:

$$L(\{\beta\}) = \sum_{i=1}^{n} \sum_{k=1}^{K} 1(Y_i = k)\, log p_k(X_i)$$

Softmax coding: Softmax coding is a way to model a multiclass response variable without choosing a

baseline class. Instead of comparing each class to a reference, we treat all K classes symmetrically. But the softmax coding is used extensively in some areas of the machine learning literature. In the softmax coding, rather than selecting a baseline class, we treat all K classes symmetrically, and assume that for $k = 1, \dots, K$:

$$Pr(Y = k|X) = p_k(X) = \frac{e^{\beta_{k0}+\beta_{k1}X_1+\dots+\beta_{kp}X_p}}{\sum_{l=1}^{K} e^{\beta_{l0}+\beta_{l1}X_1+\dots+\beta_{lp}X_p}}$$

Thus, rather than estimating coefficients for K − 1 classes, we actually estimate coefficients for all K classes.

The log-likelihood formula is the same; only the definition of $p_k(X)$ changes.